A Simplified Model for California Grapevine Leaf Water Potential Mapping at the Field Scale Based on a Machine Learning Approach

This repository and another one (LWP_Vineyard_Features) support a peer-reviewed journal paper published in *Irrigation Science* (A machine learning framework for California vineyard water status monitoring using sUAS Imagery and short-term meteorological data) showing a simplified model for California vineyard leaf water potential mapping. A subtitle or the main title is below.
In this repository, we provided:

1. `Input_data`, a folder that contains demo data. `Demo_INput_TIR.tif` is the temperature image (in Celsius) obtained from the AggieAir sUAS. `Demo_Input_VNIR.tif` is the multi-spectral image (red, green, blue, and near-infrared).
2. `main_program.ipynb` is the main program, which is a simplified model from the research **A machine learning framework for California vineyard water status monitoring using sUAS Imagery and short-term meteorological data**.
3. `xgb_tt.pkl` is the trained machine learning model (using the XGBoost approach). The required inputs are listed in the research paper, and we also list them below.
    - T_a, air temperature in Celsius at 2 m above ground level.
    - T_c, canopy temperature in Celsius.

Reference

Gao, R., Alsina, M. M., Torres-Rua, A. F., Hipps, L., Kustas, W. P., Anderson, M., ... & Dokoozlian, N. (2026). A machine learning framework for California vineyard water status monitoring using sUAS Imagery and short-term meteorological data. Irrigation Science, 44(3), 60. https://doi.org/10.1007/s00271-026-01102-8 Gao, R., Torres-Rua, A., & Alsina, M. M. (2025). A Simplified Model for California Grapevine Leaf Water Potential Mapping at the Field Scale Based on a Machine Learning Approach (v0.0.1). Zenodo. https://doi.org/10.5281/zenodo.15885590
Gao, R., & Torres-Rua, A. (2025). Feature Extraction from the High-resolution AggieAir Images for Leaf Water Potential Estimation in California Vineyards (Initial). Zenodo. https://doi.org/10.5281/zenodo.16730652

Citation

If you use this repository in your work, please consider following reference/DOIs:

BibTeX:

```
@misc{gao2025lwpis,
  author        = {Rui Gao, Maria Mar Alsina, Alfonso Torres-Rua, La
  title         = {A machine learning framework for California viney
  year          = {2026},
  publisher     = {Irrigation Science},
  doi           = {https://doi.org/10.1007/s00271-026-01102-8},
  url           = {https://link.springer.com/article/10.1007/s00271-
}
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@misc{gao2025lwpmap,
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  title         = {A Simplified Model for California Grapevine Leaf 
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  doi           = {10.5281/zenodo.15885589},
  url           = {https://doi.org/10.5281/zenodo.15885589}
}
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Contact info

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